GATE

Computer Science & IT

Computer Networks

Introduction

DPP: 1

Q1 The protocol data unit (PDU) for the transport layer in the Internal stack is
(A) Segment (B) Datagram
(C) Message (D) Frame

Q2 Which of the following functionalities must be implemented by a transport protocol over and above the network protocol?
(A) Multiplexing and Demultiplexing
(B) Segmentation
(C) Node to Node connectivity
(D) End to End Connectivity

Q3 Matching the following:

	List-I		List-II
P.	IP	1.	Application layer
Q	UDP	2.	Transport layer
R	FTP	3.	Data link layer
S	CSMA	4.	Network layer

(A) P-4, Q-2, R-3, S-1
(B) P-4, Q-3, R-1, S-3
(C) P-3, Q-2, R-1, S-3
(D) P-4, Q-2, R-1, S-3

Q4 Assume that source host and destination host are connected through three intermediate routers. Determine how many times each packet has to visit the data link layer during a transmission from source host to destination host.

Q5 Which of the following application layer protocol uses always TCP as the transport protocol?
(A) SMTP (B) FTP
(C) UDP (D) IP

Q6 Which of the following transport layer protocol is used to support web browsing?
(A) HTTP (B) IP
(C) TCP (D) SMTP

Q7 How many bits in the port number field of a network application process?

Q8 A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 10111110111, then the input bit-string is
(A) 10111110111
(B) 1011111111
(C) 101111100111
(D) 1011111001110

Q9 A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the input bit-string is 10111110111, then the output bit-string after stuffing is
(A) 10111110111 (B) 1011111111
(C) 101111100111 (D) 1011111001110

Q10 A bit-stuffing based framing protocol uses an 6-bit delimiter pattern of 011110. If the input bit-string is 10111110111, then count the number of stuffed bits in the output bit-string. [Except start and end frame delimiter]

Q11 A bit-stuffing based framing protocol uses an 5-bit delimiter pattern of 01110. If the output bit-string is 10110110101101, then count the number

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of stuffed bits in the output bit-string. [Except start and end frame delimiter]

Q12 In a data link protocol, the frame delimiter flag is given by 01110. Assuming that bit stuffing is employed, the transmitter sends the data sequence 11001110011 as

(A) 11001110011 (B) 110111011
(C) 11000110100110 (D) 1100011100011

Q13 In a data link protocol, the frame delimiter flag is given by 01110. Assuming that bit stuffing is employed, the data sequence received by receiver is : 110011010110, identify correct data sequence at transmitter end (before transmit).

(A) 11001101011 (B) 110111011
(C) 11000110010110 (D) 1100011001011

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Answer Key

Q1	(A)
Q2	(A, B, D)
Q3	(D)
Q4	8~8
Q5	(A, B)
Q6	(C)
Q7	16~16
Q8	(B)
Q9	(C)
Q10	2~2
Q11	3~3
Q12	(C)
Q13	(B)

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Hints & Solutions

Q1 Text Solution:

Layer	PDU
Application	Message
Transport	Segment
Network	Datagram
Data Link Layer	Frame

Q2 Text Solution:

Node to Node Connectivity → Data link layer

Q3 Text Solution:

IP $\rightarrow$ Network Layer

UDP $\rightarrow$ Trasport Layer

FTP $\rightarrow$ Application Layer

CSMA $\rightarrow$ Data Link Layer

Q4 Text Solution:

$1 + (3 \times 2) + 1 = 8$

Q5 Text Solution:

Application layer protocol uses TCP as the transport protocol:

- SMTP
- FTP

Q6 Text Solution:

TCP, a transport layer protocol is used to support web browsing.

Q7 Text Solution:

16 bits in the port number field of a network application process.

Q8 Text Solution:

Output string :1 0 1 1 1 1 1 0 1 1 1

Input string : 10 1 1 1 1 1 1 1 1

$\xleftarrow[\text{Output sting}]{\text{Input string}}$ Transmitter

Output string= stuffing in input string

Q9 Text Solution:

Input : 1 0 <u>1 1 1 1 1</u> 0 1 1 1

Output: 1 0 1 1 1 1 1 0 0 1 1 1

Q10 Text Solution:

Stuff bit pattern = 0 1 1 1 1 0

Input = 1 0 <u>1 1 1</u> 1 1 0 1 1 1

Output = 1 0 1 1 1 0 1 1 0 1 1 1 0

Q11 Text Solution:

Stuff bit pattern (5 bit) = 0 1 1 1 0

Output = 1 0 1 1 0 1 1 0 1 0 1 1 0 1

Q12 Text Solution:

Input : 1 1 0 0 1 1 1 0 0 1 1

Output : 1 1 0 0 0 1 1 0 1 0 0 1 1 0

Q13 Text Solution:

Output : 1 1 0 0 1 1 0 1 0 1 1

Input : 1 1 0 1 1 1 0 1 1

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